WoSM Academy

MAT231-F27 · CALCULUS II

First-order differential equations

Separable and linear equations

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Lecture 23 · Unit 23 · Fall 2027

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Last time

- Vector fields; gradient fields; flow lines $\mathbf{r}' = \mathbf{F}(\mathbf{r})$.
- Line integrals $\int_C f\, ds$ and $\int_C \mathbf{F} \cdot d\mathbf{r}$; the fundamental theorem $\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A)$.

Flow lines are solutions of differential equations. The last three units of the module are about such equations: today one unknown function and its first derivative; then second derivatives (Lecture 24) and systems (Lecture 25).

This lecture

01 Differential equations

02 Separable equations

03 First-order linear equations

04 Summary and practice

Reading: Strang & Herman, *Calculus, Volume 2*, Sections 4.1–4.5.

01

Differential equations

What is a differential equation?

An equation relating an unknown function and its derivatives. Its **order** is the highest derivative that appears.

- $y' = ky$: growth or decay at a rate proportional to the amount.
- $P' = rP\,(1 - P/K)$: growth limited by a capacity K (the logistic equation).
- $T' = -k\,(T - A)$: Newton's law of cooling, in a room at temperature A.
- $m\,x'' = -kx$: a mass on a spring (second order, Lecture 24).
- $\mathbf{x}' = A\mathbf{x}$: a system of equations (Lecture 25).

A **solution** on an interval I is a differentiable function that satisfies the equation at every point of I. Equations with one independent variable are **ordinary** (ODEs); with several, **partial** (PDEs).

Initial value problems

A first-order **initial value problem**: $y' = f(t, y),\ y(t_0) = y_0$.

> **Theorem (Existence and uniqueness)**
>
> If f and $\partial f/\partial y$ are continuous near (t_0, y_0), the problem has exactly one solution on some interval around t_0.

- **Non-uniqueness** without the hypothesis: $y' = 3y^{2/3}$, $y(0) = 0$ has the solutions $y = 0$ and $y = t^3$. Here $\partial f/\partial y = 2y^{-1/3}$ blows up at $y = 0$.
- **Only on some interval**: $y' = y^2$, $y(0) = 1$ has the solution $y = \frac{1}{1-t}$, which blows up at $t = 1$.

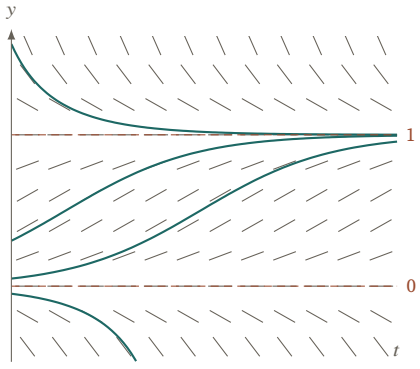
Direction fields

At each point (t, y), the equation $y' = f(t, y)$ prescribes a **slope**. Draw short segments with these slopes: solutions are the curves tangent to them.

For $y' = y(1 - y)$:

- $y = 0$ and $y = 1$ are constant solutions (**equilibria**);
- solutions starting between them rise towards 1;
- above 1 they fall towards 1; below 0 they fall away.

$y = 1$ is **stable**, $y = 0$ is **unstable**.



Euler's method

Follow the direction field in small steps of length h:

$$t_{k+1} = t_k + h, \qquad y_{k+1} = y_k + h\, f(t_k, y_k).$$

Example ($y' = y$, $y(0) = 1$; the exact value $y(1) = e \approx 2.71828$)

$y_{k+1} = (1 + h)\, y_k$, so the estimate of $y(1)$ is $(1 + h)^{1/h}$:

$$\begin{aligned} h = 0.5: &\quad 1.5^{2} &&= 2.25 \\ h = 0.1: &\quad 1.1^{10} &&\approx 2.5937 \\ h = 0.01: &\quad 1.01^{100} &&\approx 2.7048 \end{aligned}$$

The error is roughly proportional to h. Gradient descent (Lecture 15) is Euler's method for the flow lines of $-\nabla f$, with step $h = \eta$.

02

Separable equations

Separation of variables

An equation is **separable** if $y' = g(t)\,h(y)$.

Method

1. Constant solutions: every y_0 with $h(y_0) = 0$ gives the solution $y = y_0$.
2. Elsewhere, divide by $h(y)$ and integrate: $\displaystyle\int \frac{dy}{h(y)} = \int g(t)\,dt + C$.
3. Solve for y if possible, and use the initial condition to find C.

Why it works: if $H' = 1/h$, then by the chain rule $\frac{d}{dt}H(y(t)) = \frac{y'(t)}{h(y(t))} = g(t)$. So $H(y(t)) = \int g\,dt + C$.

Exponential growth and decay

$y' = ky$: $\displaystyle\int \frac{dy}{y} = \int k\,dt$ gives $\ln|y| = kt + C$, so

$$y(t) = y_0\, e^{kt}$$

(this also covers the constant solution $y = 0$).

- $k > 0$: growth, doubling time $\frac{\ln 2}{k}$. $k < 0$: decay, half-life $\frac{\ln 2}{|k|}$.

> **Example (Newton's law of cooling, $T' = -k(T - A)$)**
>
> $T(t) = A + (T_0 - A)\, e^{-kt}$. A cup at 90°C in a room at 20°C is at 60°C after 10 minutes: $e^{-10k} = \frac{4}{7}$. It reaches 40°C when $e^{-kt} = \frac{2}{7}$: $t = 10\,\frac{\ln(7/2)}{\ln(7/4)} \approx 22.4$ minutes.

The logistic equation

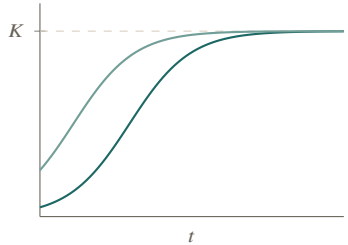
$P' = rP\left(1 - \frac{P}{K}\right)$, with $r, K > 0$. Separate, and use partial fractions:

$$\int \left(\frac{1}{P} + \frac{1}{K - P}\right) dP = \int r\, dt,$$

so $\ln \frac{P}{K-P} = rt + C$ and

$$P(t) = \frac{K}{1 + A\, e^{-rt}}, \qquad A = \frac{K - P_0}{P_0}.$$

The **sigmoid** $\sigma(x) = \frac{1}{1+e^{-x}}$ of logistic regression is the case $r = K = 1$: $\sigma' = \sigma(1 - \sigma)$.



Growth slows as P approaches the capacity K.

A solution can blow up

Example ($y' = t\,y^2$, $y(0) = 1$)

$\displaystyle\int \frac{dy}{y^2} = \int t\,dt$: $-\frac{1}{y} = \frac{t^2}{2} + C$, and $y(0) = 1$ gives $C = -1$. So

$$y(t) = \frac{2}{2 - t^2}, \qquad -\sqrt{2} < t < \sqrt{2}.$$

- The formula makes sense for $|t| > \sqrt{2}$ too, but that is a different branch: the solution of the initial value problem lives on the interval containing $t_0 = 0$.
- Nothing in $f(t, y) = ty^2$ warns of this: nonlinear equations can blow up in finite time. Linear equations cannot (next).

03

First-order linear equations

Linear equations and integrating factors

A first-order **linear** equation: $y' + p(t)\,y = q(t)$.

Integrating factor

Let P be an antiderivative of p, and $\mu(t) = e^{P(t)}$. Then $\mu' = p\mu$, so

$$(\mu y)' = \mu y' + p\mu y = \mu q, \qquad \text{hence} \qquad y(t) = \frac{1}{\mu(t)}\left(\int \mu(t)\,q(t)\,dt + C\right).$$

Example ($y' + 2y = e^{-t}$, $y(0) = 0$)

$\mu = e^{2t}$: $(e^{2t}y)' = e^{t}$, so $e^{2t}y = e^{t} + C$ and $y = e^{-t} + Ce^{-2t}$. $y(0) = 0$ gives $C = -1$: $y = e^{-t} - e^{-2t}$.

Another example

Example ($t\,y' + y = t^2,\ t > 0$)

In standard form $y' + \frac{1}{t}\,y = t$: $p = \frac{1}{t}$, $P = \ln t$, $\mu = t$. Indeed the left side of the original equation is already $(ty)'$:

$$(ty)' = t^2, \qquad ty = \frac{t^3}{3} + C, \qquad y = \frac{t^2}{3} + \frac{C}{t}.$$

- If p and q are continuous on an interval, the solution exists on the whole interval: linear equations do not blow up.
- Here $p = \frac{1}{t}$ is continuous only for $t > 0$ (or $t < 0$); the solutions with $C \neq 0$ blow up at $t = 0$.

Example: a mixing tank

Example (A mixing tank)

A tank holds 100 L of water. Brine with 0.5 kg of salt per litre flows in at 2 L/min; the mixture flows out at 2 L/min. With $S(t)$ the salt in kg:

$$S' = \underbrace{0.5 \cdot 2}_{\text{in}} - \underbrace{\frac{S}{100} \cdot 2}_{\text{out}} = 1 - \frac{S}{50}, \qquad S(0) = 0.$$

$\mu = e^{t/50}$, and $S(t) = 50\big(1 - e^{-t/50}\big)$: the salt approaches 50 kg, the amount at the incoming concentration.

The structure of the solutions

The general solution of $y' + py = q$ is

$$y = y_p + C\, y_h,$$

where y_p is one particular solution and $y_h = e^{-P(t)}$ solves the homogeneous equation $y' + py = 0$.

- The map $y \mapsto y' + py$ is **linear**. So the difference of two solutions of $y' + py = q$ solves $y' + py = 0$.
- The same structure as for $A\mathbf{x} = \mathbf{b}$ in Linear Algebra: one particular solution plus the null space.
- Superposition: if y_1 solves the equation with q_1 and y_2 with q_2, then $y_1 + y_2$ solves it with $q_1 + q_2$.

Second-order linear equations (Lecture 24) and linear systems (Lecture 25) have the same structure, with a null space of dimension 2, or n.

04

Summary and practice

Summary

- An initial value problem $y' = f(t, y)$, $y(t_0) = y_0$ has a unique solution near t_0 if f and f_y are continuous. Without that, uniqueness can fail; solutions can blow up.
- Direction fields show solutions at a glance; Euler's method follows them numerically.
- **Separable** $y' = g(t)h(y)$: $\int dy/h(y) = \int g\,dt$, plus the constant solutions. Growth and decay, cooling, the logistic equation.
- **Linear** $y' + py = q$: multiply by $\mu = e^{\int p}$, then $(\mu y)' = \mu q$. Solutions: particular + homogeneous.

Practice

1. Solve $y' = -2ty$, $y(0) = 3$.
2. Solve $y' = 1 + y^2$, $y(0) = 0$. On which interval does the solution exist?
3. Solve $y' = e^{t-y}$, $y(0) = 1$.
4. A population follows $P' = P(1 - P/10)$ with $P(0) = 1$. When does it reach 5?
5. Solve $y' - y = t$, $y(0) = 0$.
6. Solve $t\,y' + 2y = 1$ for $t > 0$.
7. Use Euler's method with $h = 0.25$ to estimate $y(1)$ for $y' = y$, $y(0) = 1$.
8. Show that $y' = 2\sqrt{|y|}$, $y(0) = 0$ has more than one solution. Which hypothesis of the theorem fails?

Answers

1. $\ln|y| = -t^2 + C$: $y = 3e^{-t^2}$.
2. $\arctan y = t$: $y = \tan t$, on $(-\frac{\pi}{2}, \frac{\pi}{2})$.
3. $e^y\,dy = e^t\,dt$: $e^y = e^t + e - 1$, $y = \ln(e^t + e - 1)$.
4. $P = \frac{10}{1+9e^{-t}} = 5$ when $e^{-t} = \frac{1}{9}$: $t = \ln 9 \approx 2.20$.
5. $\mu = e^{-t}$: $(e^{-t}y)' = te^{-t}$, $y = -t - 1 + Ce^t$, $C = 1$: $y = e^t - t - 1$.
6. $\mu = t^2$: $(t^2 y)' = t$, $y = \frac{1}{2} + \frac{C}{t^2}$.
7. $1.25^4 \approx 2.441$ (against $e \approx 2.718$).
8. $y = 0$, and $y = t^2$ for $t \geq 0$ (with $y = 0$ for $t < 0$). $\partial f/\partial y = 1/\sqrt{y}$ (for $y > 0$) is unbounded near $y = 0$.

Next time

Lecture 24 — Second-order linear equations (Unit 24)

- $ay'' + by' + cy = 0$: the characteristic equation and its three cases
- Oscillations: springs, damping, and Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$
- Forced oscillations: undetermined coefficients, and resonance

Reading: Strang & Herman, *Calculus, Volume 3*, Sections 7.1–7.3.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.